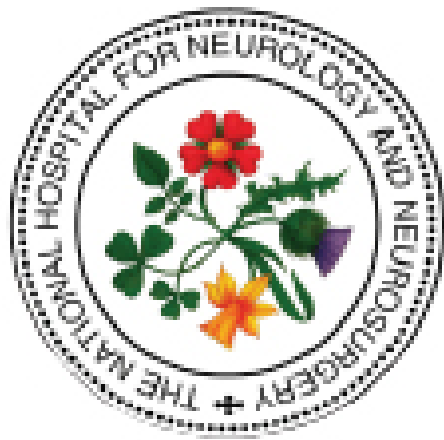




THE NATIONAL HOSPITAL FOR NEUROLOGY AND NEUROSURGERY

BLOWN PUPIL PATHWAY

This pathway is only to be applied to patients at The National Hospital for Neurology and Neurosurgery

A “**blown pupil**” is a medical emergency indicating trans-tentorial herniation and brainstem compression. This can be caused by a variety of surgical reversible pathologies such as haematoma, hydrocephalus, large volume infarct or oedema surrounding tumour or infection. Rapid access to surgical evacuation of extra-axial haematomas significantly improves outcomes and is the most important factor in improving prognosis. Progression to definitive surgical intervention must occur within ‘the golden hour’ to prevent secondary brain injury and irreversible brainstem ischaemia.

MINIMUM ACTIVATION CRITERIA

To meet minimum activation criteria the patient must have:

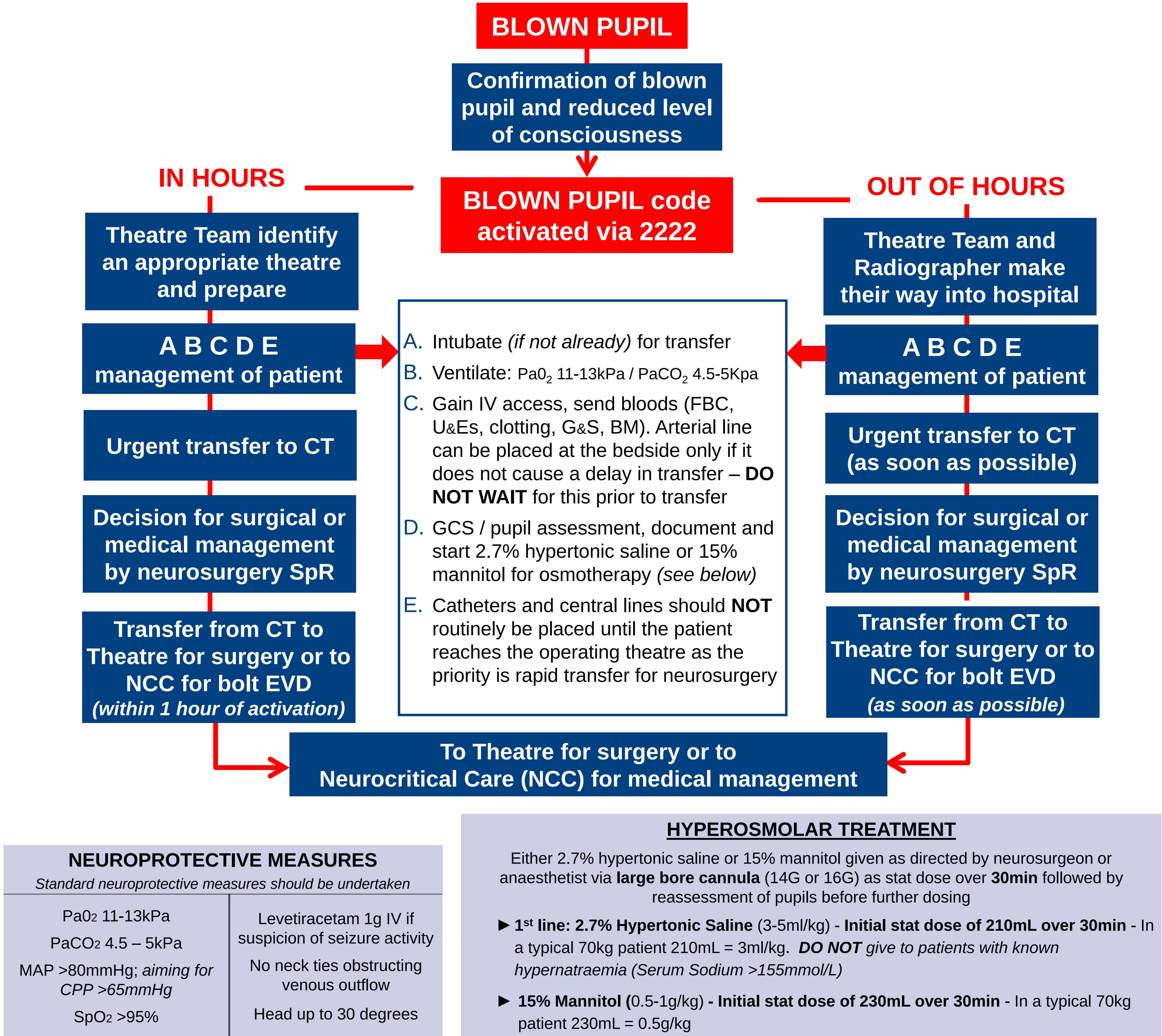
1. Impaired consciousness (GCS <15)
2. At least one pupil that is both FIXED (completely unreactive to light) and DILATED (>5mm)
3. A reasonable clinical suspicion that pupillary dilatation is secondary to intracranial mass effect

ACTIVATION

- The pathway can be activated by any member of staff trained in making above clinical assessment of pupils and medical context
- For patients being transferred from another hospital this will be activated by the on call Neurosurgery Registrar, including dissemination of estimated time of arrival (ETA) and clinical details
- The pathway is activated via the **SWITCHBOARD OPERATOR** through the emergency number **2222**
- On speaking to the operator the activator should request a ‘**Blown Pupil**’ call is activated - this will send a fast bleep/phone call to the following:

TEAM MEMBERS

- On call Neurosurgery Registrar
- On call Neurosurgery SHO
- First On call Anaesthetic Registrar
- On call Radiographer
- Theatre Coordinator (*in hours*) or on call Theatre Team (*out of hours*)
- Neurocritical Care Outreach
- Neurocritical Care Nurse-in-Charge
- On call Radiology Registrar
- Porter
- Neurosurgery Consultant (*Daytime hours only*). Neurosurgery registrar to liaise with consultant out of hours
- Neurocritical Care Consultant (*Daytime hours only*). Anaesthetic registrar to liaise with consultant out of hours
- Anaesthetic Floor Consultant / On call consultant (*Daytime hours only*). Anaesthetic registrar to liaise with consultant out of hours



THE NATIONAL HOSPITAL FOR NEUROLOGY AND NEUROSURGERY

BLOWN PUPIL PATHWAY

HYPEROSMOLAR THERAPY

Calculations based on lowest recommended doses

SODIUM CHLORIDE 2.7% BOLUS	
WEIGHT (kg)	mL based on 3mL/kg
40	120
45	135
50	150
55	165
60	180
65	195
70	210
75	225
80	240
85	255
90	270
95	285
100	300
110	330
>120	360

MANNITOL 15% BOLUS	
WEIGHT (kg)	mL based on 0.5g/kg
40	130
45	150
50	165
55	180
60	200
65	215
70	230
75	250
80	265
85	280
90	300
95	315
100	330
110	365
>120	400